Network Analysis
Causes of Death across Countries

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Background

Data (Sources, Integration, Cleaning)

EDA

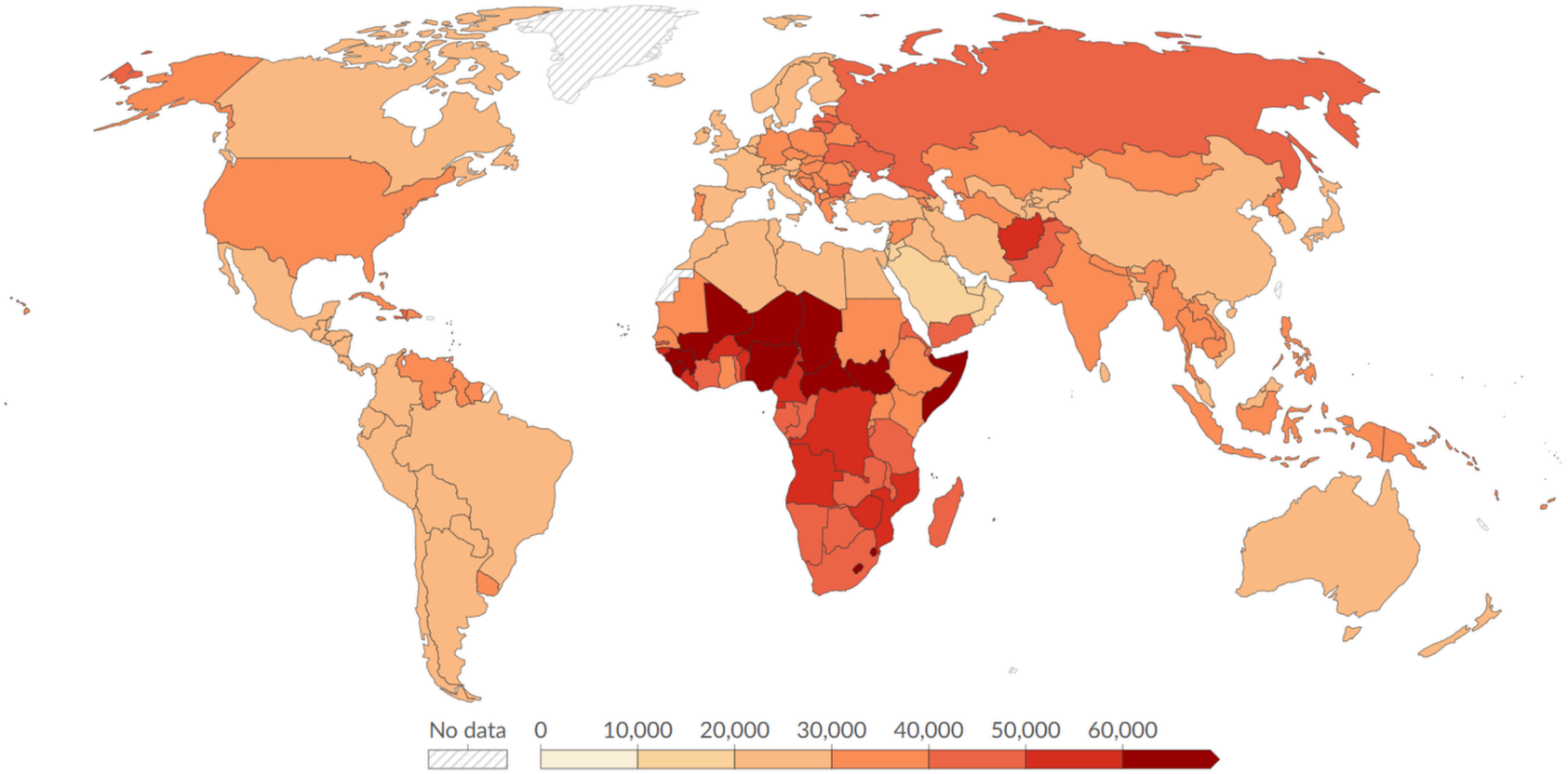
Network Analysis

Network Evaluation

Conclusion and Findings

BACKGROUND

BURDEN OF DISEASE (2019)



Why do countries exhibit different disease mortality patterns?

Global differences in disease burden

Economic and healthcare inequalities

Potential connections between diseases

RESEARCH OBJECTIVES

WHAT LIES BEHIND



How are different causes of death associated across countries?

Explore shared mortality patterns



Which disease associations remain after accounting for economic development?

Control for GDP-related patterns



Can we identify meaningful disease communities with health indicators?

Control for GDP-related patterns

DATA SOURCES

COLLECTION & INTEGRATION



WHO GHE

Cause-specific age-standardized death rates



World Bank

Economic, demographic and development indicators



WHO GHO

Health systems, risk factors and coverage



OWID

Human Development Index

Data Integration Pipeline

ISO3 standardization · Country-year alignment · Merge

Integrated Dataset

185 countries · 2000–2023 · 304 columns

DATA INTEGRATION PIPELINE

FROM RAW DATA TO AN ANALYSIS-READY DATASET

Integrated Dataset

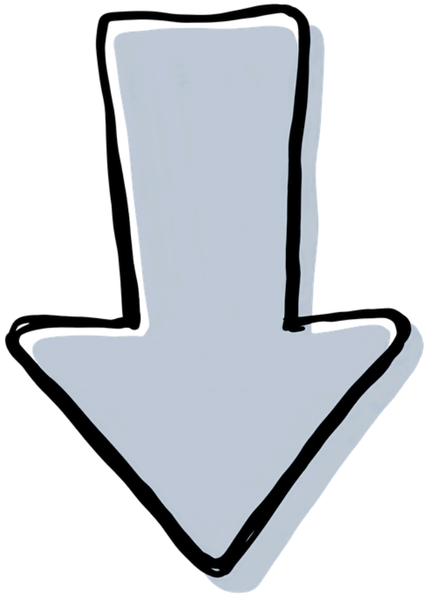
185 countries · 2000–2023 · 304 columns

1.TEMPORAL ALIGNMENT

2019 mortality;
latest indicators from 2013–2019

2.FEATURE SELECTION

Remove aggregate,
sparse and unsuitable variables



3.MISSING VALUE TREATMENT

20% country threshold; income-group
and global median imputation

4.DISTRIBUTION TRANSFORMATION

log1p for selected
highly right-skewed variables

Final Analytical Dataset

182 countries · 139 causes · 71 indicators · 0 missing values

FINAL DATASET

STRUCTURE & FEATURES

182
Countries
Observations

139
Causes of Death
Mortality Variables

71
Health Indicators
Additional Variables



What does one observation contain?

Country	Mortality Variables (139)	Health Indicators (71)
Germany	Heart disease, Malaria, Diabetes	GDP, Doctors, Obesity, HDI

2019 cross-sectional dataset

Mortality: 2019 age-standardized death rates
Indicators: latest available 2013-2019
No remaining missing values

139 Causes of Death	
Cadiovascular Disease	Ischaemic heart disease, stroke, ...
Cancers	Lung cancer, breast cancer,...
Infectious Disease	Tuberculosis, malaria, HPV/HIV, ...

71 Health Indicators	
Healthcare Resources	Doctors, nurses, hospitalbeds, ...
Healthcare Financing	Health Expenditure, out-of-pocket spending
Risk Factors	Obesity, smoking, alcohol, hypertension

Exploratory Data Analysis

Network Analysis
of Causes of Death
across countries

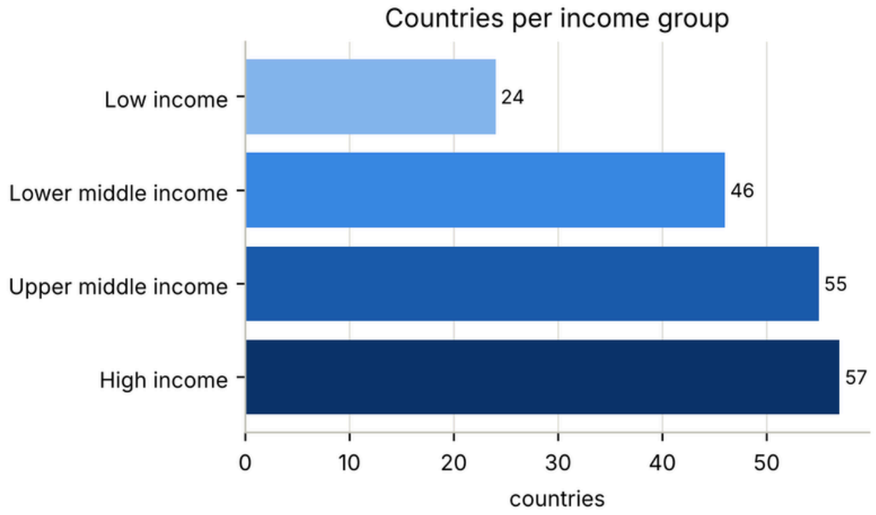
Data at a Glance

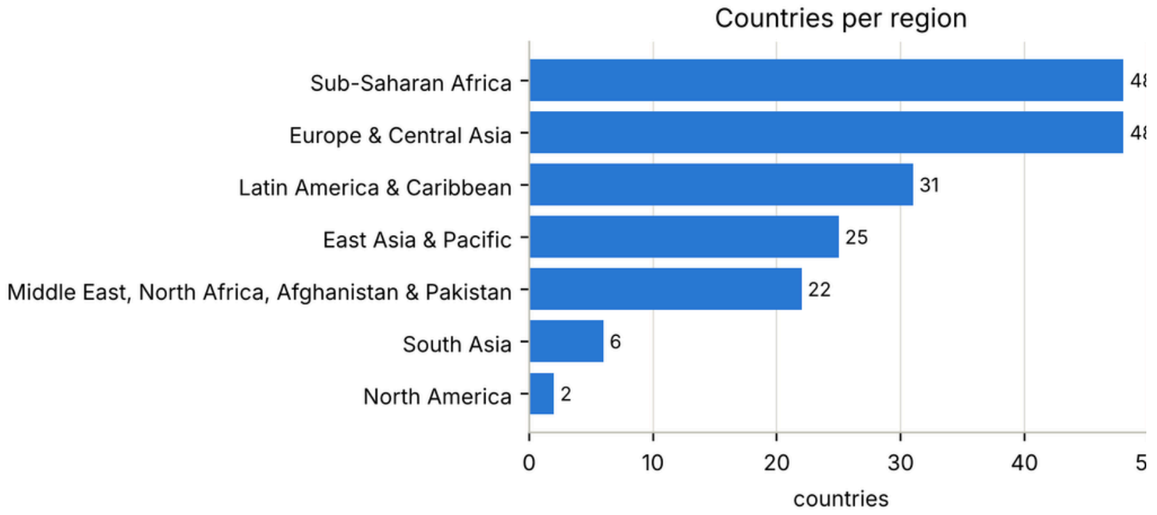
2019 cross-section, one row per country

182
countries

71
indicators

139
causes of death

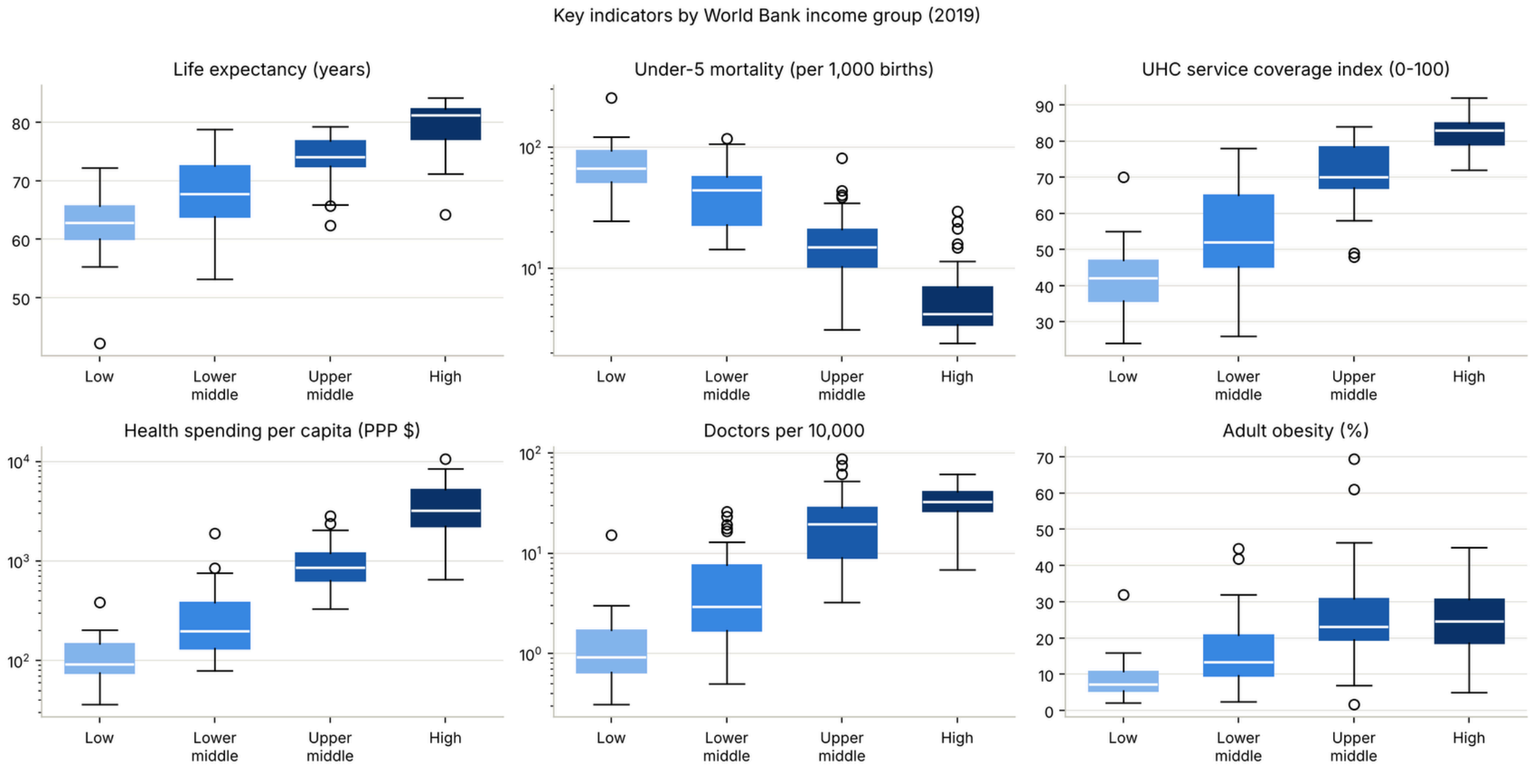




Light blue is low income, dark blue is high income

A Steep Income Gradient

Key health indicators by World Bank income group



18 more years

Life expectancy, rich vs. poor countries

35 times more resources

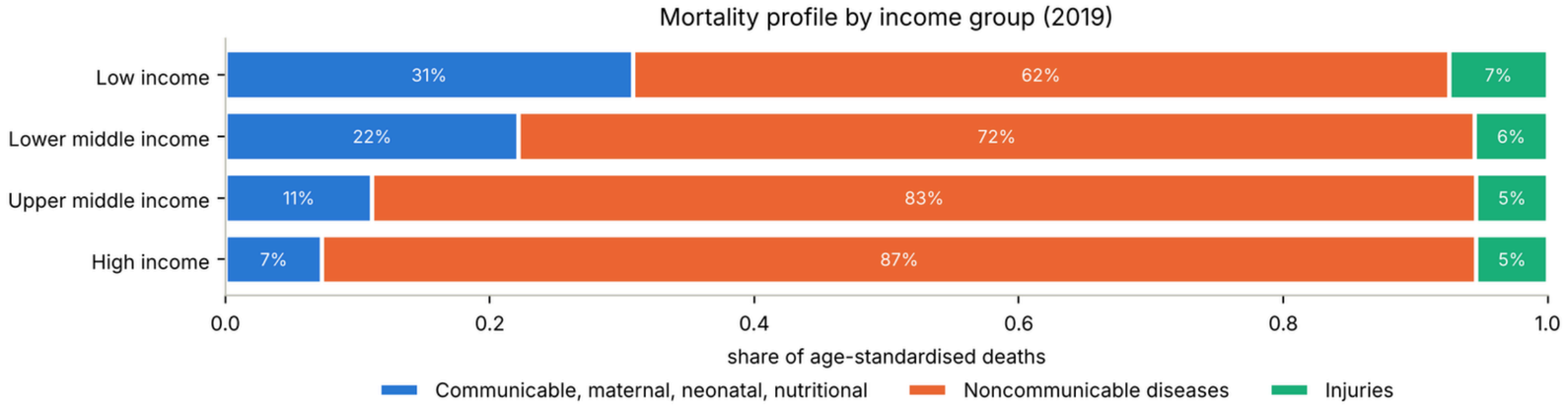
Health spending and doctors per person

Same income, different results

Coverage varies widely within each group

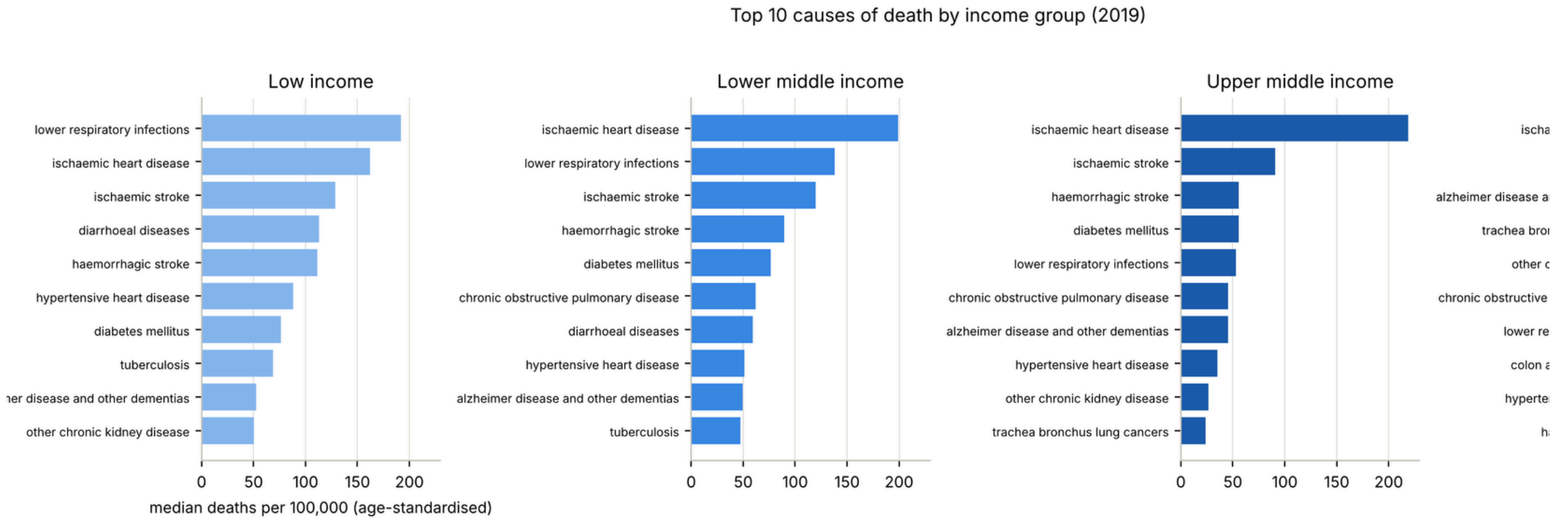
What People Die Of

Mortality profile and top 10 causes by income group



Share of deaths

	Low income	High income
Infectious	31%	7%
Chronic	62%	87%



Double burden

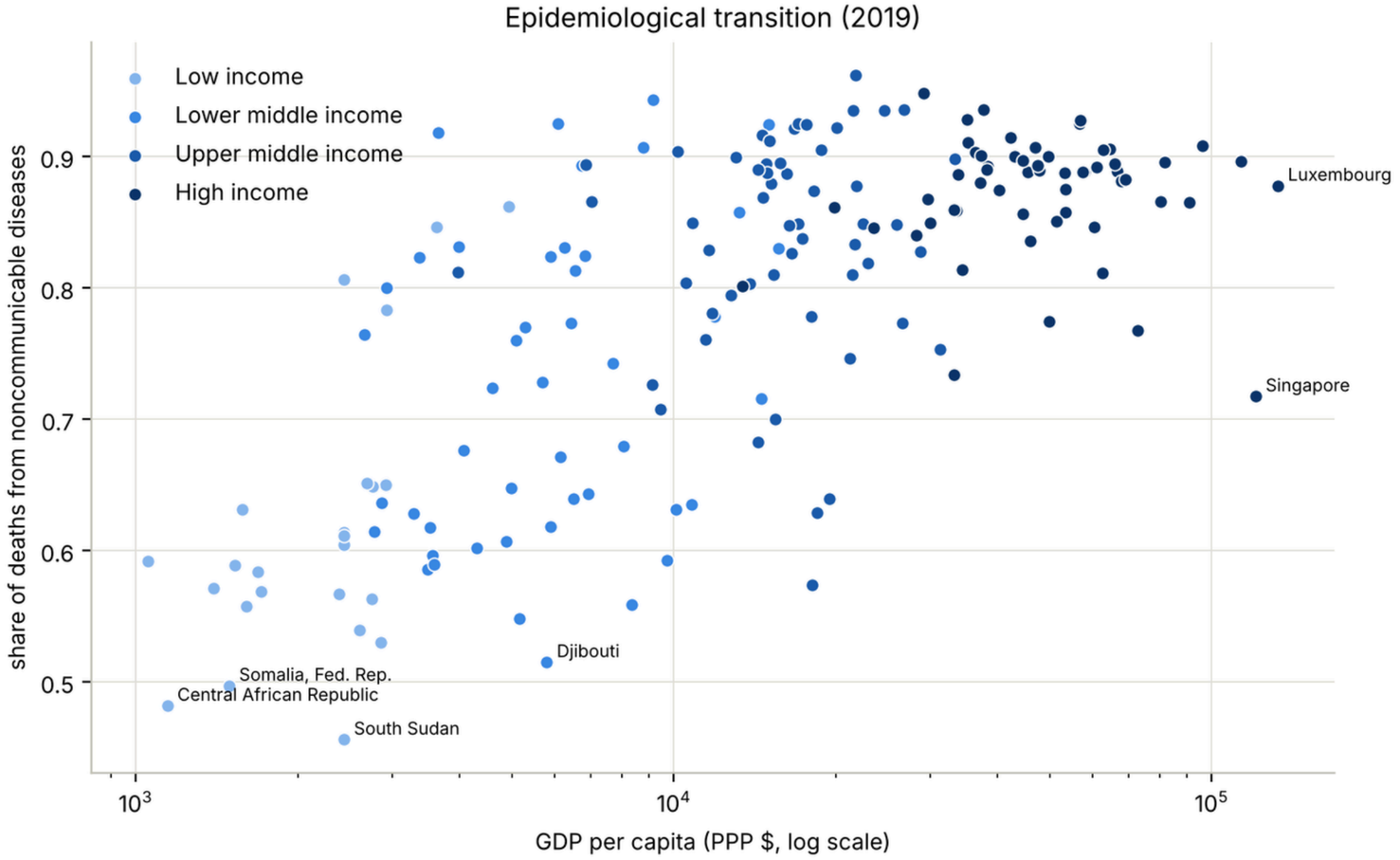
Chronic diseases dominate even in the poorest countries

Heart disease everywhere

A top-two cause of death in every income group

Epidemiological Transition

Share of noncommunicable deaths vs. GDP per capita



Richer, more chronic

The share levels off above about 20,000 $ per person

Conflict matters too

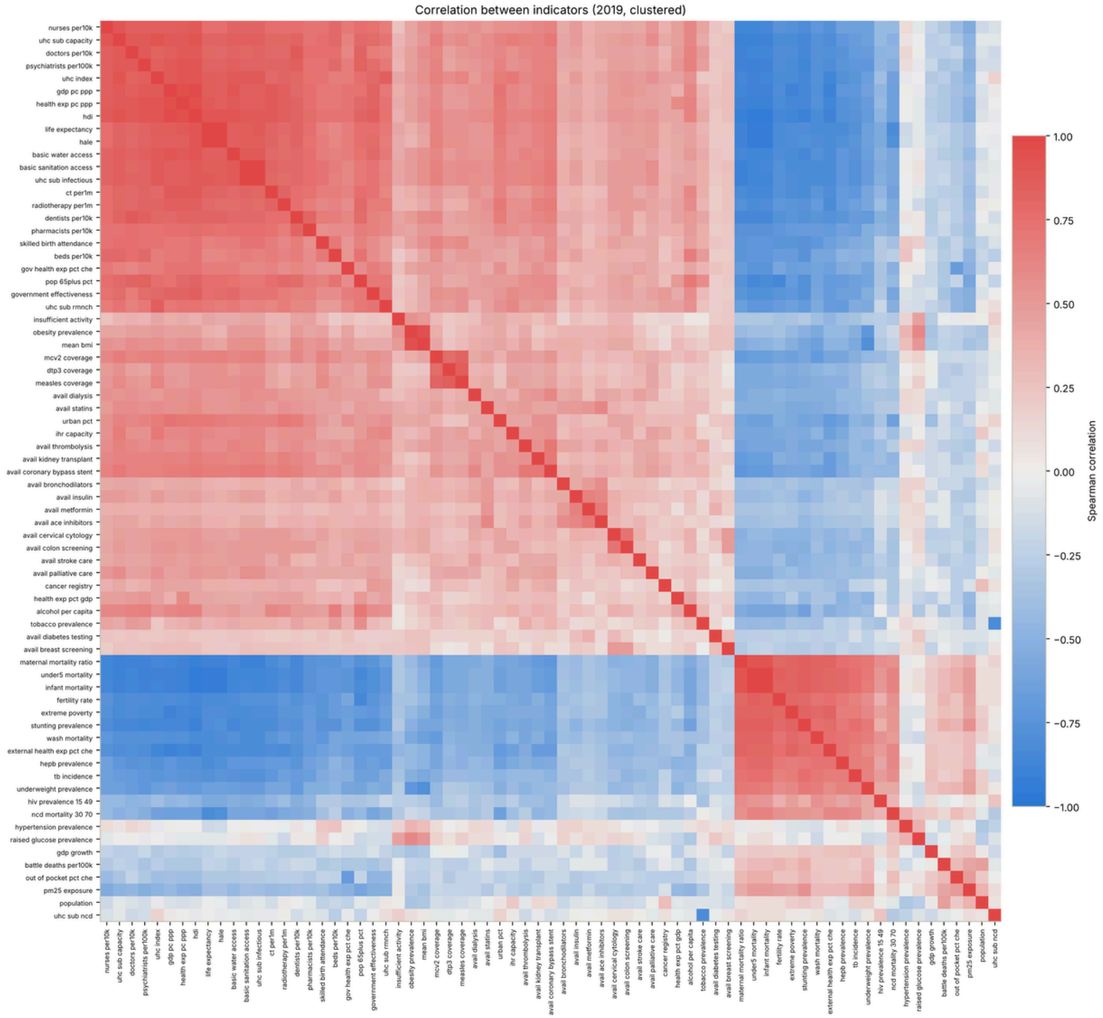
South Sudan, CAR and Somalia sit at the bottom

Singapore stands out

Pneumonia is its leading cause of death

One Dominant Development Axis

Spearman correlation between all 71 indicators (clustered)



Rise together

Doctors, spending, GDP, coverage and life expectancy

Move the opposite way

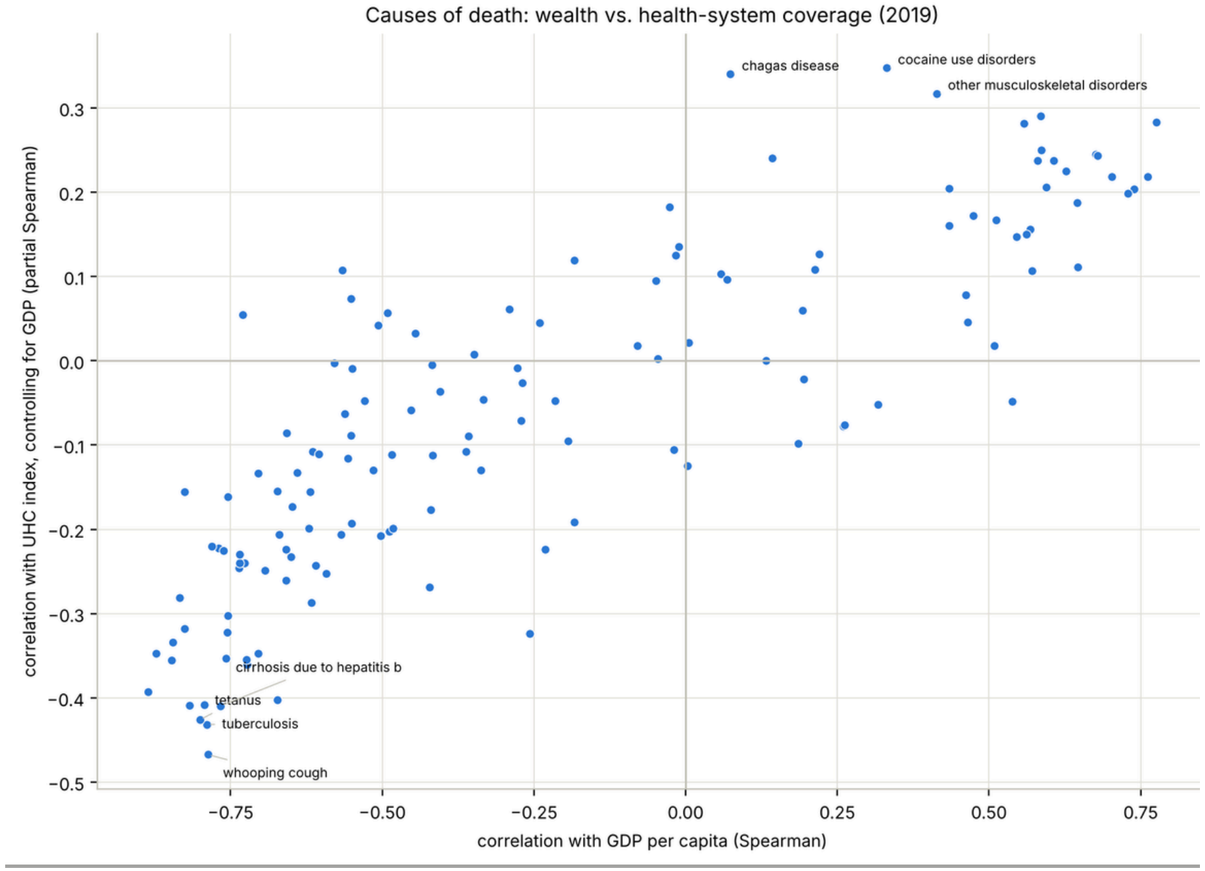
Child mortality, fertility, poverty and TB

24 near-duplicate pairs

Such as under-5 and infant mortality, so we keep one of each

Health Systems Matter Beyond Wealth

Each dot is a cause of death, compared between countries with similar income



Fewer deaths where coverage is higher

Whooping cough, TB, tetanus, measles, maternal deaths and malaria

Exactly what vaccines, TB treatment and birth care prevent

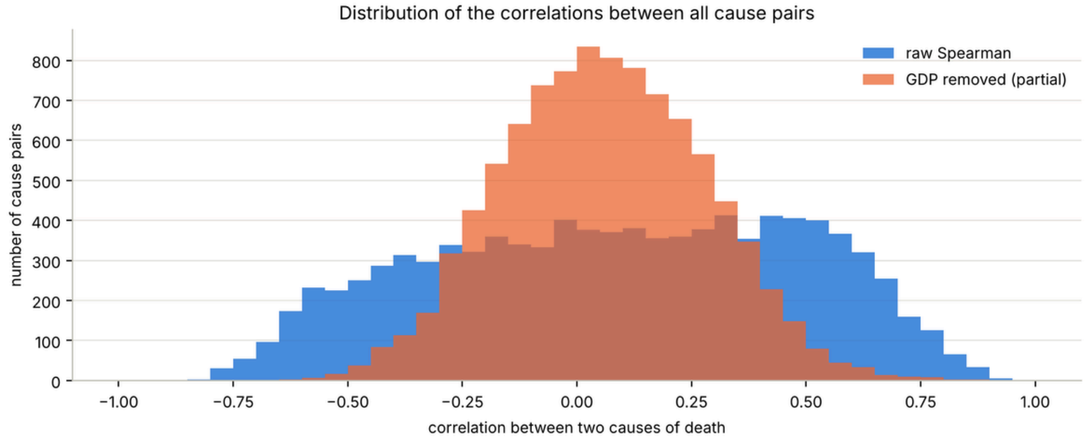
Read with care

Association, not causation

Strong systems also diagnose and record more deaths

Most Correlation Is Just Wealth

How disease links change once GDP is removed



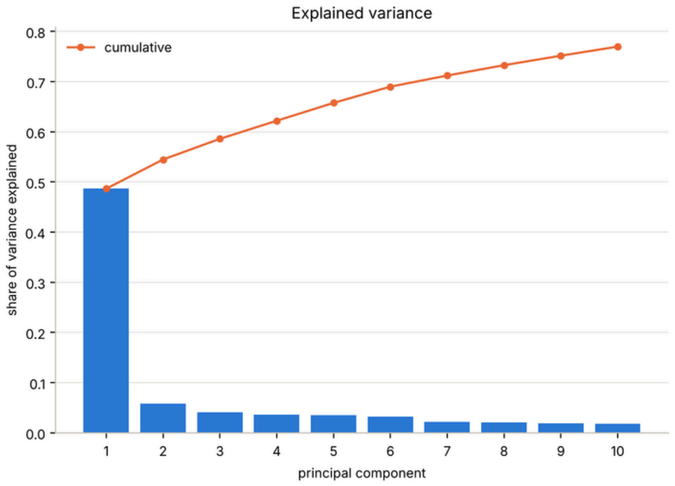
27%

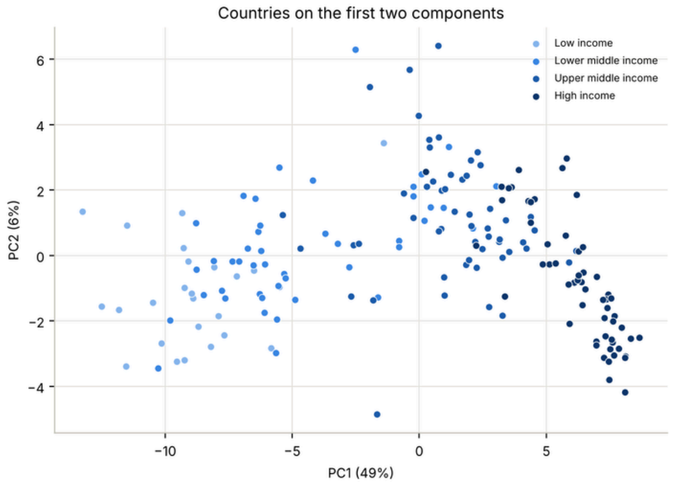
before

2%

after removing GDP

of disease pairs are strongly linked





One axis explains 49%

Development drives most of the differences between countries

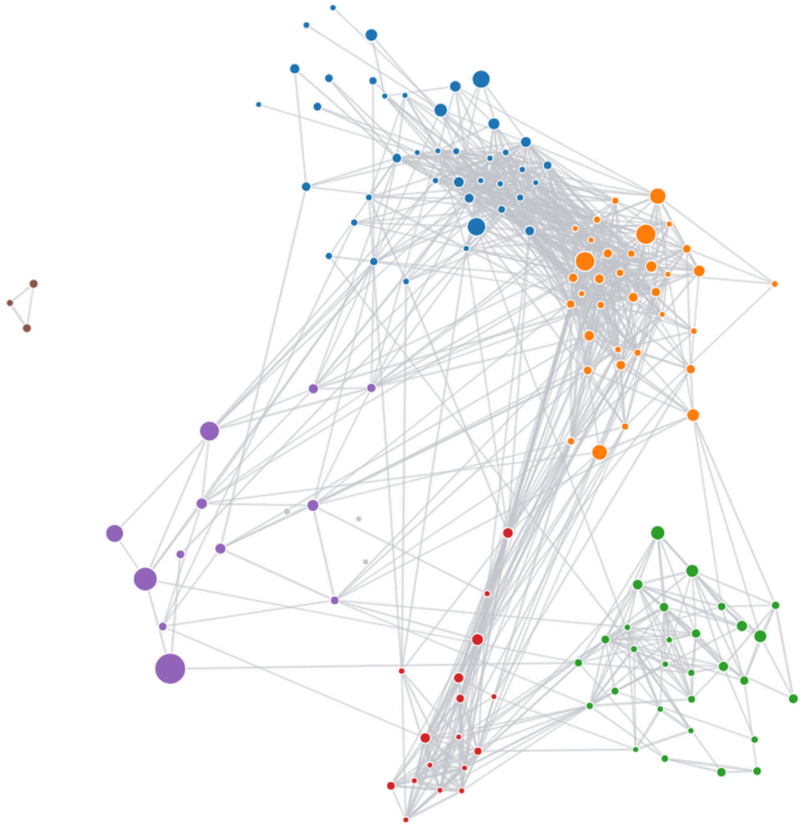
So our network removes wealth first

Network Analysis

Network Analysis
of Causes of Death
across countries

NETWORK SELECTION

COMPARISON OF BOTH NETWORKS



CO-OCCURRENCE



DIRECT LINK
(GRAPHICAL LASSO)

NETWORK SELECTION

COMPARISON OF BOTH NETWORKS

CO-OCCURRENCE

Advantages

STABLE

FINDS **16/22** KNOWN PAIRS

SHOWS **LARGER** GROUPS

Disadvantages

INDIRECT LINKS (HIV - DROWNING)

LARGER, **COARSE** COMMUNITIES

	Co-occurrence	Direct link
Links	918.00	443.00
Communities	6.00	11.00
Modularity	0.46	0.58
Stability (ARI)	0.57	0.40

DIRECT LINK (GRAPHICAL LASSO)

Advantages

REMOVES **INDIRECT LINKS**

MORE **DETAILED**

DATA DECIDES SPARCITY

Disadvantages

LOSES SOME **REAL LINKS**

WEAKER LINKS

NETWORK SELECTION

GRAPHICAL LASSO - GAUSSIAN GRAPHICAL MODEL



ONLY LINKS CAUSES IF RELEVANT AFTER ACCOUNTING FOR ALL CAUSES

USES **CORRELATION MATRIX** AND **INVERSE**

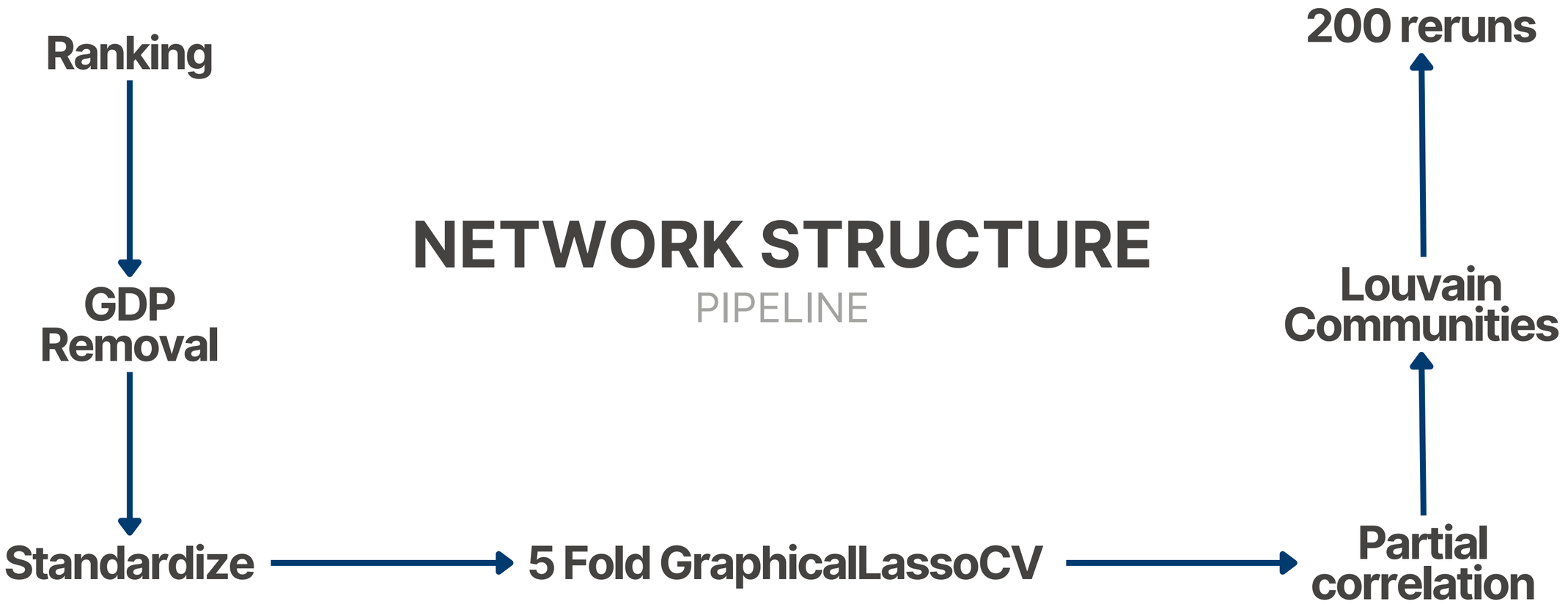
LASSO INTRODUCES PENALTY OF **NON-ZERO LINKS**

WEAK LINKS ARE PUSHED TO ZERO

5-FOLD VALIDATION

NETWORK STRUCTURE

PIPELINE



WHY DID WE GO FOR AN UNSUPERVISED APPROACH?

WE DO NOT ASK FOR HOW MUCH, BUT FOR WHAT GOES TOGETHER

Data Availability

The Data is sparse.

Model difficulties

One model per condition would have been needed.

Result Interpretation

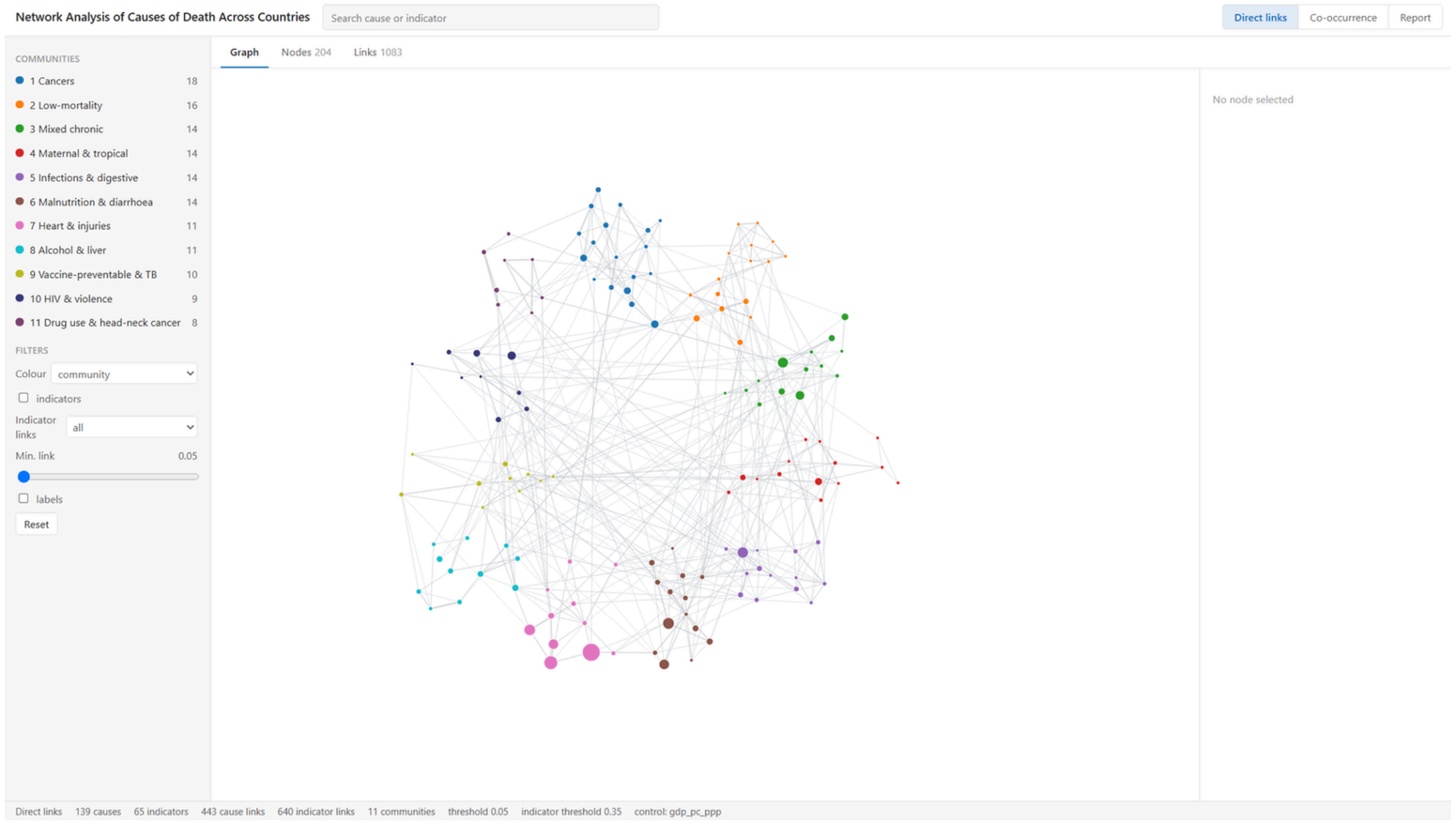
Does not awnser the research question sufficiently enough.

The Network

Network Analysis
of Causes of Death
across countries

THE NETWORK

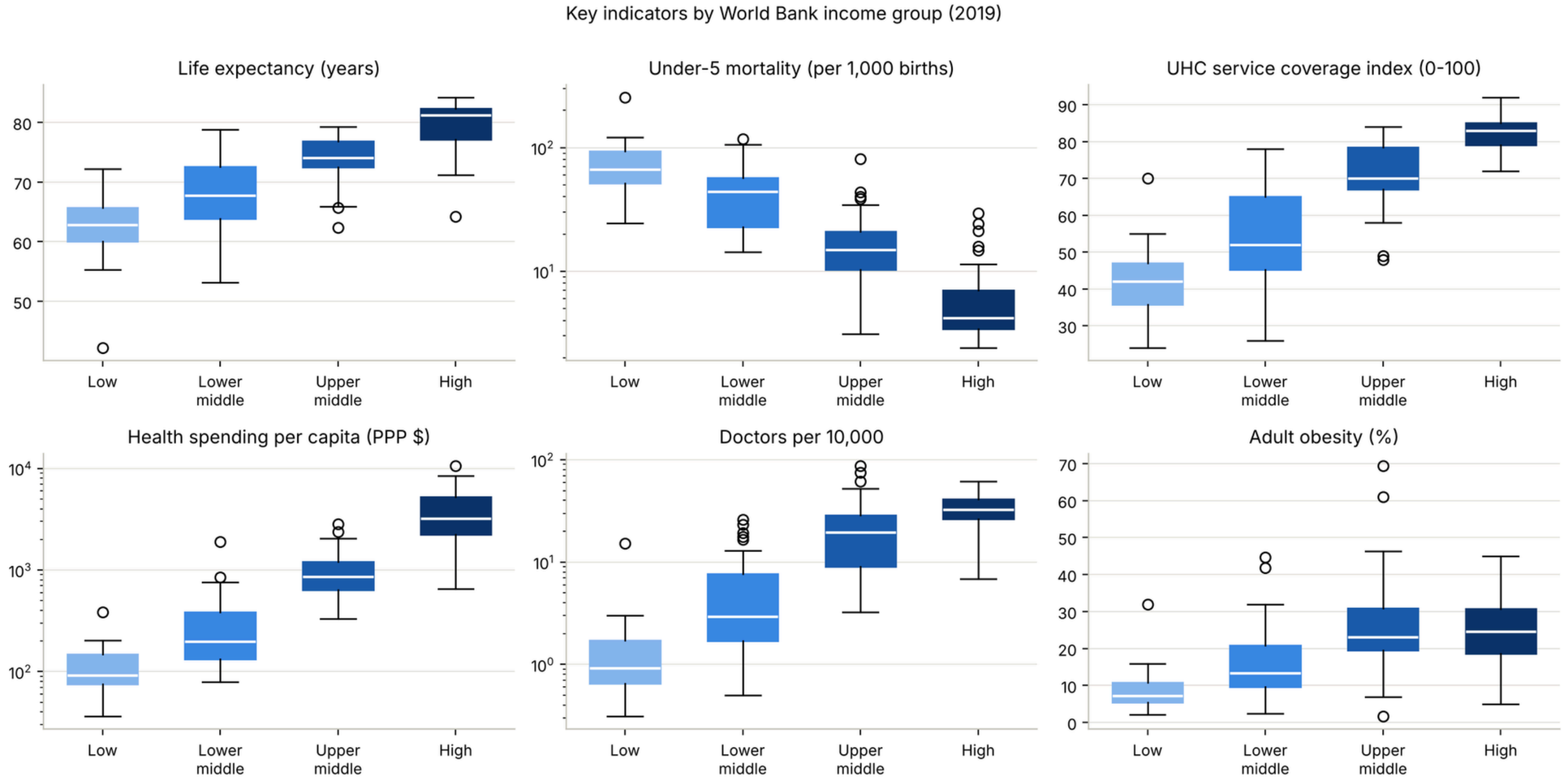
DEMO



data.yannhinsche.de

THE NETWORK

DEMO



WEALTH HIDES LINKS

STRUCTURE & FEATURES



23/29
Expected
Links

3 Stable Groups
Cancers, malnutrition & vacine-preventable

Links after adjusting data for GDP

	Raw	GDP adjusted
Obesity - Diabetes	0.01	0.37
Obesity - Malaria	-0.60	-0.44

Sanitation, doctors and infectious-disease
Affect mortality the most overall

LIMITATIONS

WITH THE APPROACH

ASSOCIATION BETWEEN COUNTRIES, NOT PATIENTS

LIMITED TO 2019

GRAPHICAL LASSO REMOVES REAL LINKS

REGIONAL LEFTOVERS WITHIN FINAL DATA

What does removing wealth actually achieve?

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FUTURE RESEARCH

Causal Inference approaches to estimate how **treatment** of one disease **could affect others**

Use Links to predict how changes in **certain factors** will **affect other factors**

Identify **regions** where treatment could yield the **best results**

Thank you

Network Analysis
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across countries